



Fire Protection System Specification

ENS-D10

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VERSION HISTORY

Version	Date	Designer	Reviewed By	Description
1.0	Sep. 30, 2025	Chi Zhang	Xiangyu Ge	Initial release
1.1	Feb. 06, 2026	Zhiyi Pu	Chi Zhang	Edit Typos
1.2	Feb. 11, 2026	Zhiyi Pu	Chi Zhang	Updated general arrangement based on updated number of aerosol dispensers (Figure 1) Updated responsive actions (Table 2)
2.0	May. 21, 2026	Zhiyi Pu	Chi Zhang	Updated according to the system configuration modification

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1. DESIGN THEME

Envision Energy's fire safety strategy is rooted in the ethos of "putting prevention first and combining prevention and containment." Our approach focuses on preventing and containing thermal runaway events through a multi-tiered safety protocol covering prevention, detection, suppression, and containment. Central to our philosophy is the thorough understanding of the thermal runaway mechanisms and mitigating the risks accordingly with efficiency. Envision Energy's comprehensive fire safety design and protocols address every facet of fire prevention, detection, suppression, and containment.

Prevention by Design:

- Advanced raw material recipe and engineering to improve lithium-ion battery's intrinsic safety.
- Implementation of rigorous manufacturing management and quality control protocols to ensure exceptional cleanliness and performance uniformity throughout the lengthy battery manufacturing process.

Early Detection:

- Deployment of gas detection systems, including hydrogen, carbon monoxide sensors, to identify off-gassing concentrations and facilitate rapid response and early risk mitigation.
- Integration of a Battery Management System (BMS) with a three-layer control topology, actively monitoring and regulating cell parameters to prevent failures and maintain optimal performance.

Suppression and Containment:

- Implementation of total-flooding fire suppression systems to ensure effective and precise suppression of fire incidents.
- Adoption of fire-resistant enclosure designs to prevent thermal propagation under sustained large-scale fire conditions.

2. FIRE PROTECTION SYSTEM GENERAL ARRANGEMENT

The fire protection system detects, alarms, and suppresses thermal runaway events at early stage. Each modular battery segment is equipped with smoke sensor, heat sensor, and combustible gas sensors to swiftly identify potential risks of battery or electrical fires. The active ventilation and deflagration venting system are designed to NFPA 69 and 68.

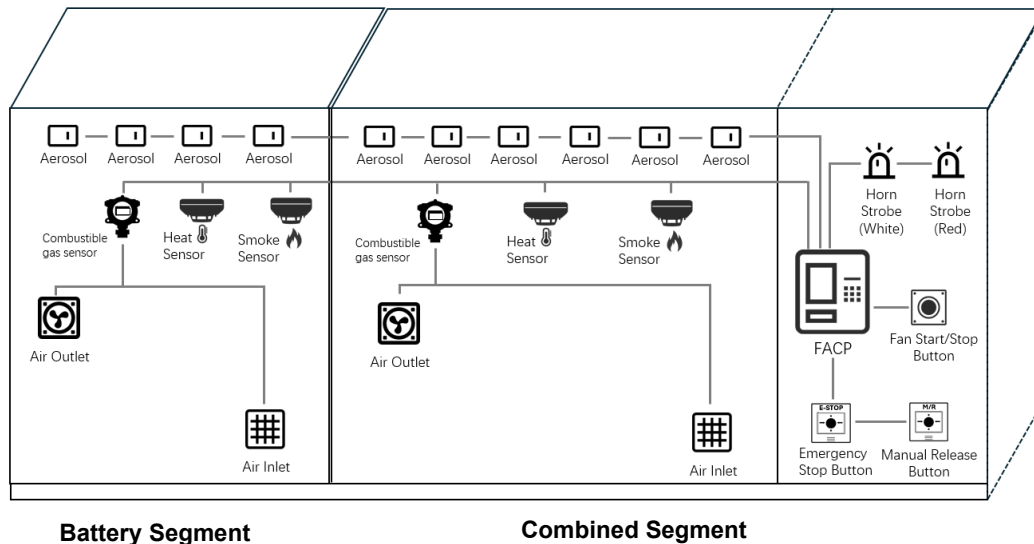


Figure 1. General arrangement of fire protection system of ENS-D10

Table 1 Key components of the fire protection system (ENS-D10)

Fire Protection System Components	Battery Segment	Combined Segment
	# per Segment	# per Segment
Smoke sensor	1	1
Heat sensor	1	1
Combustible gas sensor	1	1
Aerosol dispenser	4	6
Ventilation system	1	1
Fire alarm control panel	0	1
Manual release button	0	1
Emergency stop (for aerosol release)	0	1
Fan start/stop button	0	1
Horn strobe (Red/White)	0	2
Deflagration panel (optional)	2	3

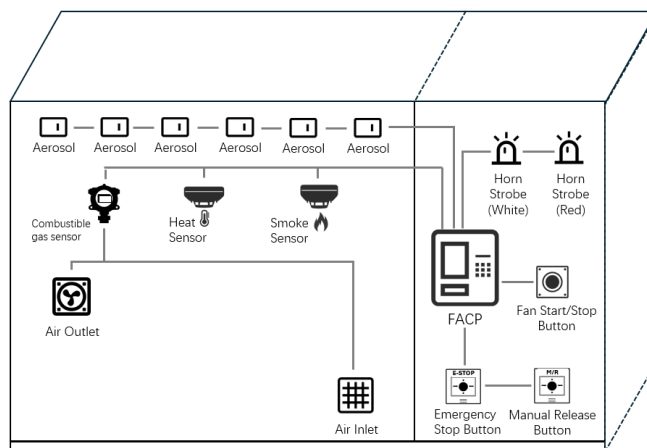


Figure 2 General arrangement of fire protection system of ENS-D6

Table 2 Key components of the fire protection system (ENS-D6)

Fire Protection System Components	ENS-D6
	# per Segment
Smoke sensor	1
Heat sensor	1
Combustible gas sensor	1
Aerosol dispenser	6
Ventilation system	1
Fire alarm control panel	1
Manual release button	1
Emergency stop (for aerosol release)	1
Fan start/stop button	1
Horn strobe	2
Deflagration panel	3

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3. FIRE ALARM SETTINGS AND CONTROL LOGIC

A two-level alarm setting is implemented for the system. The level 1 alarm is triggered by any one of the smoke sensors, temperature sensors, or combustible gas sensors within the entire system. The level 2 alarm is activated by triggering heat sensor and smoke sensor simultaneously within an individual segment.

Table 3 Fire alarm levels and responsive actions

Alarm Levels	Conditions	Responsive Actions
Level 1 Warning: A/B/C	A: Smoke sensor is triggered	Alarm bell is triggered. Charging or discharging is disabled for the entire system.
	B: Heat sensor is triggered	Alarm bell is triggered. Charging or discharging is disabled for the entire system.
	C: Combustible gas sensor in the battery compartment	For the entire system: • Alarm bell is triggered. • Charging or discharging is disabled For the segment with triggered sensor: • Active ventilation system starts up.
Level 2 Alarming: A/B	A: Smoke sensor and heat sensor in the same segment	For the entire system: • Horn strobe is triggered. • Battery charging or discharging is disabled. For the segment with triggered sensors: • Aerosol system is initiated with a 30-second delay.
	B: Aerosol is manually initiated	For the entire system: • Horn strobe is triggered. • Battery charging or discharging is disabled. • Aerosol system is initiated for all segments with a 30-second delay.
Component failure	FPS component failure	Warning message transmitted to EMS and SCADA

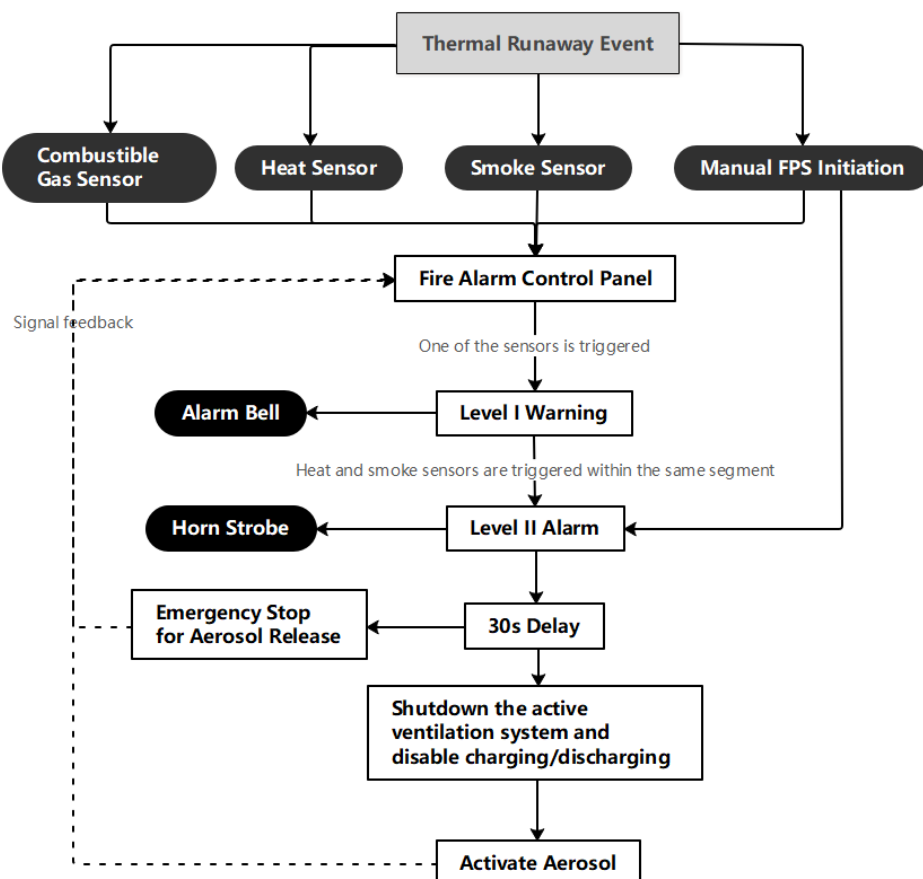


Figure 3 Fire suppression control logic

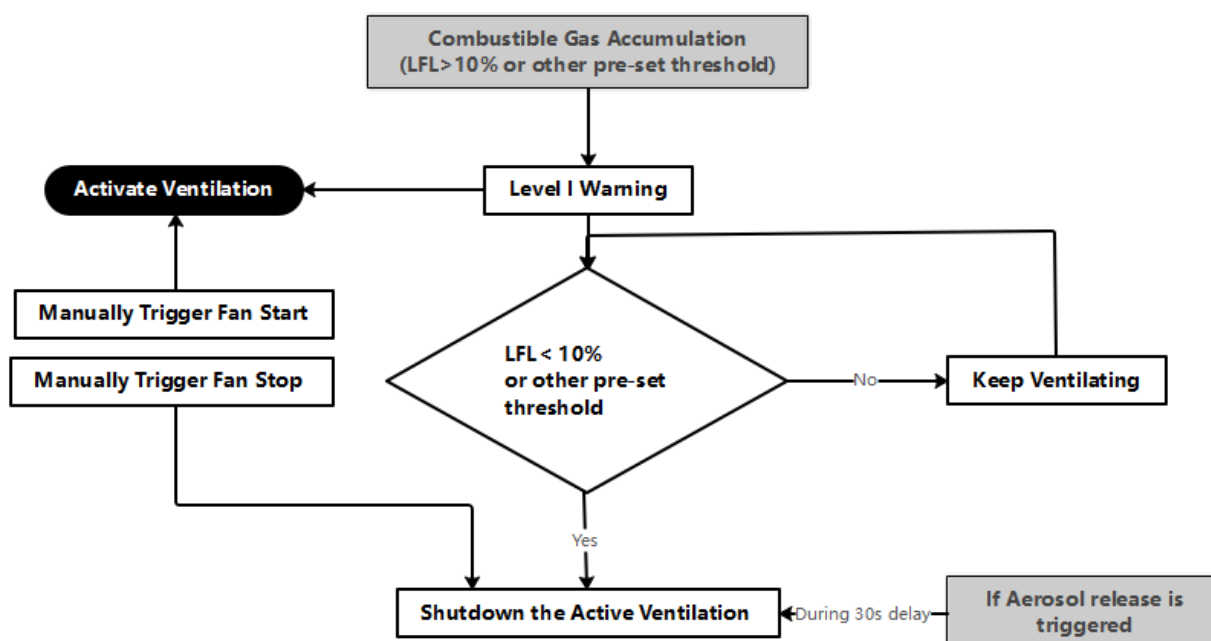


Figure 4 Active ventilation control logic

4. FIRE PROTECTION SYSTEM COMPONENTS

4.1 Heat and Smoke Sensors

The heat and smoke sensors are installed to detect temperature abnormalities and smoke release at early stage of thermal runaway events. Both heat and smoke sensors are securely mounted with anti-loosening fasteners and designed with anti-interference signal transmission capabilities.

Table 4 General specification: heat and smoke sensor

Specifications	Smoke Sensor	Heat Sensor
Sensor Type	Point detector	Point detector
Operating Voltage	10 – 30 VDC	10 – 30 VDC
Key Features	Opto-electronic measuring chamber provides excellent detection of smoke particles	Reliable response behavior for slow and rapid temperature rises
Code Compliance	EN54-7 CE, LPCB, VdS	EN 54-7 CE, LPCB, VdS

4.2 Combustible Gas Sensors

Each modular battery segment and combined segment is equipped with one combustible gas sensor. An alarming signal is generated once the gas concentration reaches 10% LFL or other pre-set thresholds.

Table 5 General specification: combustible gas sensors

Combustible Gas Sensor Specifications	
Operating Voltage	24 VDC
Current Loop	4 – 20 mA
Measuring Range	10%LEL
Code Compliance	CE, ETL

4.3 Alarm System

The alarming system consists of an alarm bell and a horn strobe. The alarm bell is triggered under level 1 fire alarm, where a single sensor is activated. The horn strobe is triggered under level 2 fire alarm, where the heat and smoke sensors are activated simultaneously.

Table 6 General specification: alarm bell and horn strobe

Specifications	Alarm Bell	Horn Strobe
Operating Voltage	24 VDC	24 VDC
Sound Output Frequency	NA	800Hz to 950Hz swept at 9Hz
Flash Rate	NA	1 per second
Sound Pressure Level	96 dB at 1m	100 dB at 1m
Code Compliance	CE, LPCB	CE, LPCB

4.4 Active Ventilation System

The active ventilation system consists of electrically actuated ventilation louvers, explosion-proof fans, and a central controller. The active ventilation system is linked with the hydrogen and carbon monoxide detectors to receive start and stop commands.

Once the gas concentration is detected above the pre-set thresholds, the electric louvers are opened, and the explosion-proof fan starts to discharge until the combustible gas concentration falls below the pre-set values.

Table 7 General specification: active ventilation

General Specification	Exhaust Fan
Rated Voltage	DC 24V
Nominal Operating Airflow	1000 CFM
Protection Class	IP55
Explosion-Proof Rating	Ex-d IIC T4
Code Compliance	CE, ATEX, IEC-Ex

4.5 Deflagration Panel

A deflagration vent panel serves as a crucial safety component designed to release pressure in the event of a sudden pressure build-up, preventing potential damage to the enclosure. The deflagration panels are installed on the top of the enclosure. When the pressure within the enclosure exceeds the static opening pressure threshold (typically around 0.1 bar), the vent panel is designed to passively open to relieve the pressure. This passive deflagration mechanism minimises the risk of structural damage or catastrophic failure of the enclosure.

The sizing and design of the explosion vent panel is carried out according to NFPA 68 and EN 14797.

Table 8 General specification: deflagration panel

Deflagration General Specifications	
IP Rating	IP 55
Material	316L stainless steel
Static Activation Pressure	0.1 bar-g
Code Compliance	CE, ATEX, EN 14797, EN 14994, NFPA 68

4.6 Aerosol

Table 9 General specification: aerosol system

Aerosol System Specifications	
Operating Voltage	24 VDC
Initiating Current	$\leq 2A$
Operating Temperature	-40 – 50°C
Storage Temperature	-45 – 60°C
Operating Humidity	Relative humidity $\leq 100\%$
Code Compliance	CE (EN 15276-1&2), BSI

4.7 Fire Alarm Control Panel

The fire alarm control panel (FACP) aggregates addressable signals from sensors and responds to a potential thermal runaway event. The FACP is installed in the combined segment, supervising the sensor status throughout all modular segments with the EN 10 system platform. Backup power is supplied by batteries to sustain continuous alarming for at least 2 hours and essential communication functions for 24 hours.

Table 10 General specification: FACP

FACP Specifications	
Rate Voltage	230 Vac
Dimensions (L x H x T)	436 x 402 x 185 mm
Backup Battery	12V, 12Ah
Code Compliance	CE (EN 54-2&4, EN 12094-1)